Lesson 12 Teacher's Key: Hearing It Happen

Scratch - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Formalizing five lessons' worth of optional 'add a sound' ideas (Lessons 2, 3, 4, 6) into two mandatory new sounds (wave clear, game over), placed in the exact branches Lesson 11 already built - no new Scratch concept, just recognizing where an event already happens and adding one block there.

Level 1 - Full Worked Solution

- sfx_hit now lives directly in the touching-Bullet branch rather than a separate broadcast - simpler than the original Lesson 4 idea, matching how this branch has been restructured since Lesson 5/7.
- sfx_wave and sfx_lose are the two genuinely new sounds this week, placed at the exact two spots Lesson 11 and Lesson 8 already built: the else branch of the wave-check, and the lives-reaches-zero check.
- If a student built Lesson 6's optional sfx_win, it needs zero changes - it already lives in the correct (true-win) branch of the same nested if.

Why This Structure

- Every sound this week lives inside a branch that only executes once per actual event (a hit, a wave clear, a loss) - Scratch's clone-and-broadcast model has made every trigger point naturally one-shot since Lesson 5, so there's no risk of a sound replaying every frame.
- Keeping sfx_wave and sfx_win in opposite branches of the same if/else (rather than, say, always playing sfx_wave and separately checking whether to also play sfx_win) guarantees they can never both fire on the same event - a direct benefit of Lesson 11's branching structure.

Level 2 / Level 3 Talking Points

- Ask students who already had sfx_shoot/sfx_hit/sfx_win from earlier optional lessons to identify, out loud, exactly which two sounds are new to them today - a good way to surface who's starting fresh versus who's just adding two lines.
- For students ready for a stretch: ask what would happen if sfx_hit were accidentally placed inside the if <enemies_left = 0> block instead of right after the decrement - it would only play on the enemy that empties the wave, not every hit. A good test of whether they understand where each sound's trigger actually lives.

Common Student Mistakes

- Placing sfx_wave or sfx_win before the if/else split instead of inside their respective branches, causing both to play on every wave clear including the true win.
- Uploading a new sound file to the wrong sprite (Scratch sounds are per-sprite) and wondering why "start sound" doesn't find it.
- Forgetting that a student who skipped Lessons 2/3/4/6 entirely needs full instructions this week, not just "add two sounds" - this lesson's Handout is written to work for both cases.